Trends in Anxiety and Depression Symptoms Among U.S. Adults, 2020–2024

From Pandemic to Recovery: Analysis of COVID-19 Pandemic, Transition, and Recovery Period

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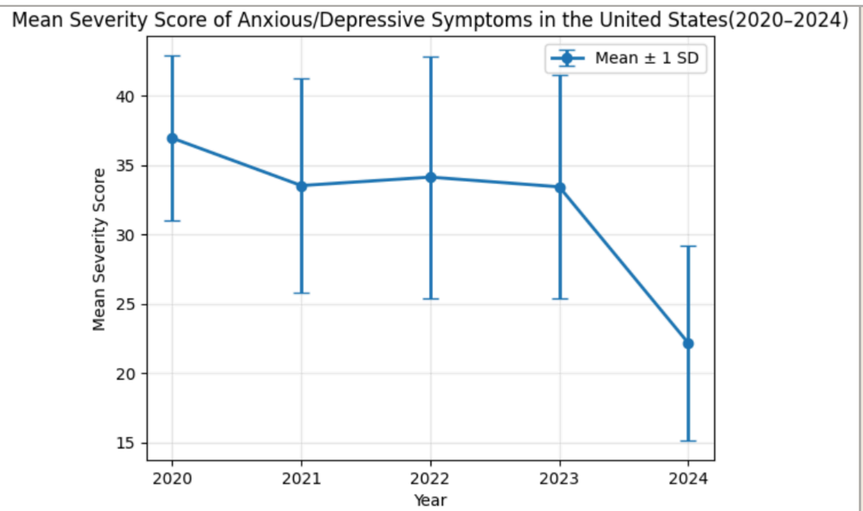


01. Introduction

The Household Pulse Survey, conducted by the National Center for Health Statistics (NCHS) in partnership with the U.S. Census Bureau, is a rapid-response, 20-minute online survey designed to capture how emerging social and economic challenges affect households nationwide. The dataset used in this study represents the mental-health component of the survey, focusing specifically on national and state-level trends in anxiety, depression and the combined symptoms during the COVID-19 pandemic (2020-2021) and through the transition (2022) and recovery periods (2023-2024).

02. Objective

To evaluate how anxiety and depression symptoms among U.S. adults changed across the COVID-19 pandemic, transitional, and recovery periods from 2020 to 2024 using validated mental-health indicators.

Research Questions

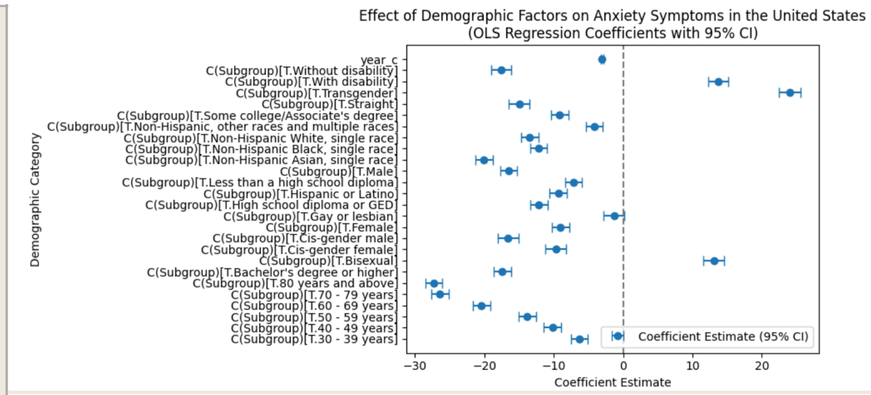
1. How have symptoms of anxiety, depression, and combined anxiety/depression changed among U.S. adults from 2020 to 2024?
2. To what extent have anxiety or depressive symptoms changed in the recovery period compared with the first year of the pandemic?
3. Which demographic groups experienced the most significant change in anxiety or depression severity levels from the pandemic period to the recovery period?
4. Are certain demographic subgroups more vulnerable to persistent anxiety/depression symptoms, especially during the recovery period?

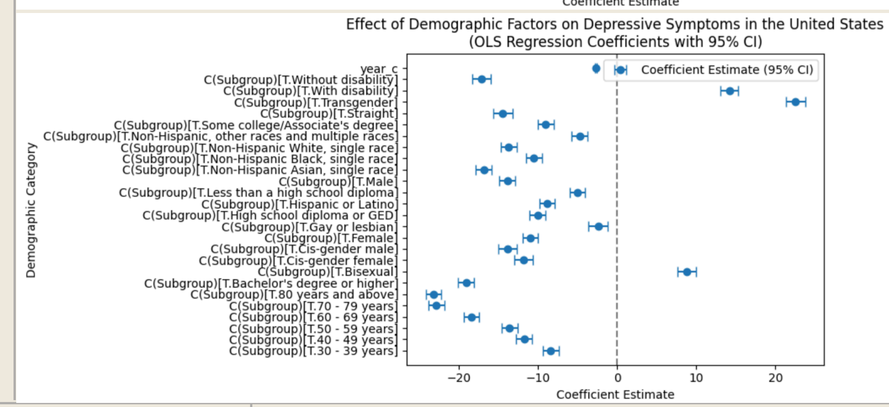
03. Methods

The data was sourced from CDC database.
The analysis was conducted in Python using packages including but not limited to pandas, numpy, matplotlib, and statsmodels to clean the data and compute annual mean severity scores and their standard deviations for anxiety, depression, and combined symptoms. The trends in different period were then compared by summarizing these statistics and visualization through a meanplot with error bars representing one SD.
A multivariable linear regression model was then fitted to estimate the independent effects of demographic variables on symptom severity of anxiety and depression, followed by generating coefficient plots using the model outputs.

04. Results

- Mean severity scores for anxious/depressive symptoms showed an overall declining trend from 2020 to 2024, with a noticeable drop during the recovery period. The peak was found in 2020 at 36.94.
- The regression model for anxiety symptoms shows a statistically significant overall decline over time (R-squared = 0.888, $p < 0.001$), indicating that anxiety levels consistently decreased from 2020 to 2024. Most subgroups displayed negative coefficients, reflecting lower anxiety levels relative to the reference group. **Bisexual individuals, transgender individuals, and people with disabilities** showed positive coefficients, reflecting elevated anxiety symptoms.
- The regression model for depressive symptoms shows a strong and statistically significant overall decline over time (R-squared = 0.909, $p < 0.001$). Similar subgroup patterns were observed as in the anxiety model, in whichbisexual individuals, transgender individuals, and people with disabilities continued to exhibit positive coefficients





05. Discussion

The findings demonstrate a statistically significant overall decline in both anxiety and depressive symptoms from 2020 to 2024, with survey year emerging as one of the strongest predictors in both models. This indicates that symptom severity have decreased consistently as the country transitioned from the peak of the COVID-19 pandemic into the recovery period, suggesting that national mental-health burden has improved over time especially during the post COVID-19 period.
However, the regression results show statistically significant disparities across demographic groups. Three subgroups consistently exhibited higher anxiety and depressive symptom severity: **bisexual individuals, transgender individuals, and people with disabilities** with a positive coefficient. These positive coefficients suggest that these populations remain at elevated risk despite the overall population-level improvement, reflecting that they may face unique stressors or structural vulnerabilities that prevent their mental-health recovery from keeping pace with the general population.
As the dataset of this analysis is a self-reported survey, it may introduce bias and measurement error. However, these results still highlight that while there is a significant general decline in mental-health symptoms over time, the improvement is not equally distributed across all subgroups.

06. Conclusion

The result shows a significant decline in anxiety and depressive symptoms among U.S. adults from the early COVID-19 pandemic through the recovery period. Despite this overall downward trend, the regression results reveal persistent and statistically significant disparities across demographic groups, with bisexual individuals, transgender individuals, and people with disabilities experiencing consistently higher symptom severity. These findings highlight the importance of continued monitoring and the need for targeted mental-health resources to support groups that have not benefited equally from the post-pandemic recovery.

year	Period	Mean Score of Depressive/Anxiety	Standard Deviation
2020	Pandemic	36.938095	5.974644
2021	Pandemic	33.518496	7.710896
2022	Transition	34.127389	8.709820
2023	Recovery	33.427040	8.054309
2024	Recovery	22.163053	7.027590